

## F2F Assignment - HTTP/HTTPS Web Proxy

### Problem Statement

You are required to write a basic HTTP proxy server that can be used as a web proxy in a web browser as well as in desktop applications like Dropbox. The proxy server should be capable of handling requests for all domains while browsing and should work seamlessly across different applications (e.g., Streaming, etc.)

### Example

Assume a user configures their browser to use your proxy server running on `localhost:8888`. When the user accesses `<http://example.com>`, the request should be routed through your proxy server, which will detect the target host and port, forward the request to `<http://example.com>`, and pipe the response back to the user's browser.

The Proxy Server must work for non-FQDN domain names as well, e.g. <http://localhost:3000>

### Submission

Please submit your source code along with instructions on how to run the proxy server. Ensure that your code is well-documented and includes comments explaining key sections.

### [ INTERNAL ] Testing & Scoring

To test the functionality of your proxy server, the following steps will be carried out:

1. Detect Target Host and Port 2 points
2. Forward Request Data to Target Website 3 points
3. Pipe Response Back to Client 2 points
  - The response received from `<http://example.com>` should match the response received by the client's browser when using the proxy server.
4. Streaming 2 points
  - A request will be made to a website that serves large files (e.g., a video file). The proxy server should stream the content back to the client without buffering issues.
5. Cross-Application Compatibility 1 point
  - The proxy server will be configured in a desktop application (e.g., Dropbox) to ensure it can

handle requests and responses seamlessly in a non-browser environment.